



PUBLIC / EXTERNAL

Compliance Whitepaper

Quenyx vOPS HUB — v1.0.0 · Document v2.0

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REVISION HISTORY

Version	Date	Notes
1.0	2026	Initial v1 pack (through Sprint 19).
2.0	2026-06-29	Aligned to v1.0.0 (Documentation Pack v2.0).

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Audience: Compliance leaders, auditors. **Thesis:** Quenyx delivers **deterministic, provenance-backed compliance intelligence** — the prerequisite for trustworthy, auditable automation in regulated environments.

1. Deterministic compliance intelligence

Quenyx's compliance engine (QCIF) is **deterministic**: the same corpus + the same evidence always produce the same gaps, the same recommendations, and the same explanations. Decisions come from **explicit rules**, not from a model's confidence. This is what makes the output defensible.

2. Official corpus

Compliance content is **imported from official source documents only**. There is no hand-authored or AI-generated control text. The `ComplianceCorpusValidator` **rejects** fake-data markers (`EXAMPLE/SAMPLE/TEST-/FAKE/LOREM/DEMO` , `lorem-ipsu`), enforcing the official-source-only model.

3. Revisions

Each release's content is captured as an **immutable Corpus Revision**; exactly one is **active**. Revisions guarantee reproducibility and enable controlled rollback. The shipped corpus is **NCA ECC-2:2024, Revision v1** (5 domains, 108 controls, 108 requirements).

4. Provenance

Every imported control/requirement carries a `source_document_id` , and every import is recorded as an **import run**. Auditors can trace any corpus element back to the official document and the import that loaded it.

5. Evidence lifecycle

Evidence is recorded per requirement/control with **types** and **statuses** (e.g. compliant, partially compliant, no evidence, expired, rejected, pending). The executive scorecard reports these as **counts**, never as invented percentages.

6. Gap rules

Gap status is **derived by rule** from evidence and requirements (e.g., a mandatory requirement with no evidence ⇒ a missing-evidence gap). Rules are explicit and enumerated in

`ComplianceReasoningRuleSet` .

7. Recommendation rules

Recommendations are produced by the same rule set (e.g., missing evidence $\Rightarrow$ "collect evidence"). Each recommendation references the **rule that produced it**, so its rationale is transparent.

8. Explainability

The explainability service answers "why" questions ("Why is this requirement non-compliant?", "Why this recommendation?") directly from the deterministic reasoning — **fired rule IDs**, findings, and citations — not from generated prose.

9. Auditability

Because the corpus is provenance-tracked, revisions are immutable, reasoning is rule-based, and answers are cited, an auditor can independently verify **every** conclusion: source $\rightarrow$ rule $\rightarrow$ finding $\rightarrow$ recommendation, each with identifiers.

10. Why this reduces hallucination risk

- AI never authors compliance facts; it **renders** deterministic outputs.
- "No citation, no answer" excludes ungrounded statements.
- RAG (when enabled) is **metadata-only with deterministic fallback** and excludes uncited chunks.
- Tenant evidence is **not embedded** by default.

The model cannot "make up" a control because the control set, its provenance, and the reasoning are fixed and external to the model.

11. Limitations

- Only **one framework (NCA ECC-2:2024)** is currently loaded.
- Evidence quality depends on what the customer supplies; Quenyx assesses, it does not collect for you.
- Quenyx models official controls but makes **no claim of certification or accreditation** by any authority.

12. Future frameworks

The authority/framework/release/revision model is framework-agnostic. Additional frameworks can be imported through the same manifest $\rightarrow$ validate $\rightarrow$ revision pipeline, inheriting the same determinism, provenance, and auditability guarantees.